Log S-fBM Random Measure Implementation

The associated Python code is `step1.py`.

Method Description

The S-fBM process $\{\omega_{H,T}(t)\}_t$ is a stationary Gaussian process constructed through the following steps:

1. **Covariance Structure**: The covariance function for $|\tau| < T$ is given by:

$$\text{Cov}[\omega_{H,T}(t), \omega_{H,T}(t+\tau)] = \frac{\nu^2}{2}\left(T^{2H} - |\tau|^{2H}\right)$$

   where $\nu^2 = \frac{\lambda^2}{H(1-2H)}$, and zero for $|\tau| \geq T$.

2. **Mean Adjustment**: The mean $\mu = -\frac{1}{2}\text{Var}[\omega_{H,T}(t)]$ ensures $\mathbb{E}[e^{\omega_{H,T}(t)}] = 1$, a key property for volatility modeling.

3. **Sampling**: Realizations are generated using multivariate normal distributions with covariance matrices computed through vectorized operations for efficiency.

Parameter Interpretations

- **Hurst Parameter (H)**:
  - Controls path roughness: Lower H (closer to 0) $\Rightarrow$ more erratic paths (anti-persistent)
  - Higher H (closer to 0.5) $\Rightarrow$ smoother trends (persistent)
  - Governs Hölder continuity: α-Hölder continuous for all $\alpha < H$ (cannot have any "sharp jumps" or discontinuities. Moreover, it satisfies the inequality:

$$|f(x) - f(y)| \leq C\|x - y\|^{\alpha},$$

    where $C > 0$ is a constant and $\alpha \in (0, 1]$.)

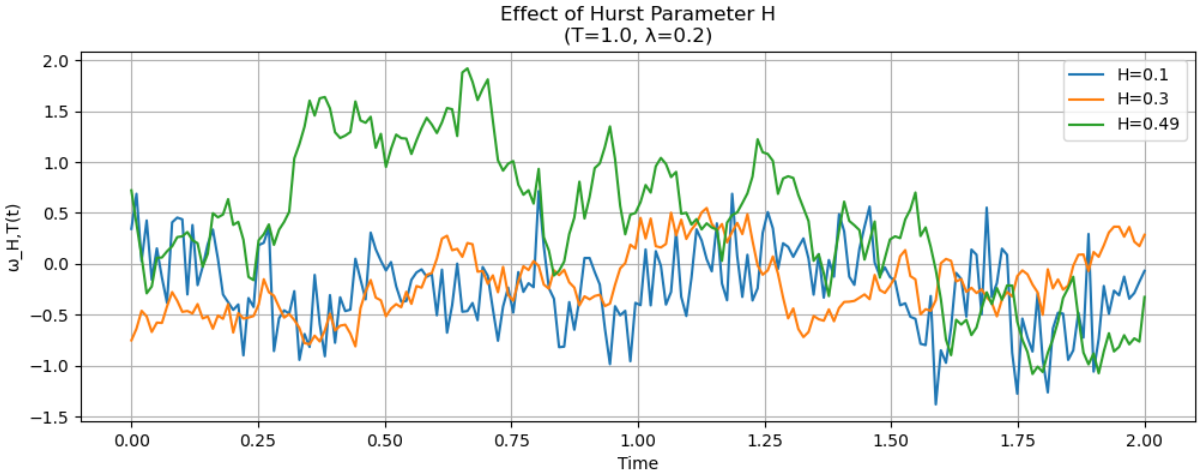


Figure 1: several H and fixed T, λ

- **Decorrelation Scale (T)**:
  - Sets correlation cutoff: $\text{Corr}(t, t+\tau) = 0$ for $|\tau| \geq T$
  - Larger T $\Rightarrow$ longer memory effects
  - As $T \to \infty$, recovers fractional Brownian motion

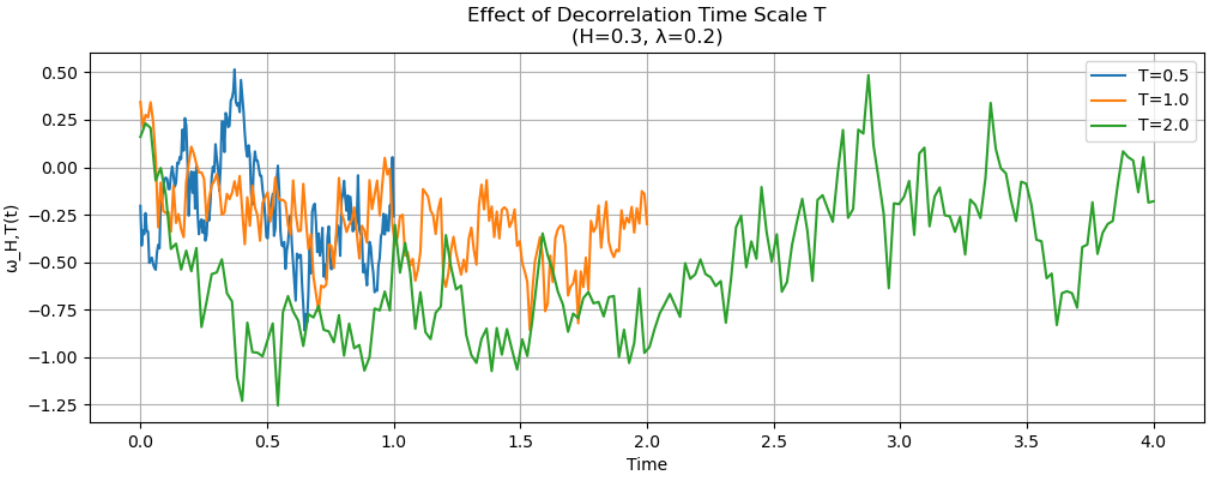


Figure 2: several T and fixed H, λ

- **Intermittency** (λ):
  - Controls volatility burst intensity via $\nu^2 = \frac{\lambda^2}{H(1-2H)}$
  - Higher $\lambda \Rightarrow$ stronger volatility clustering and extreme events
  - Linked to multifractal properties in the limit $H \to 0$

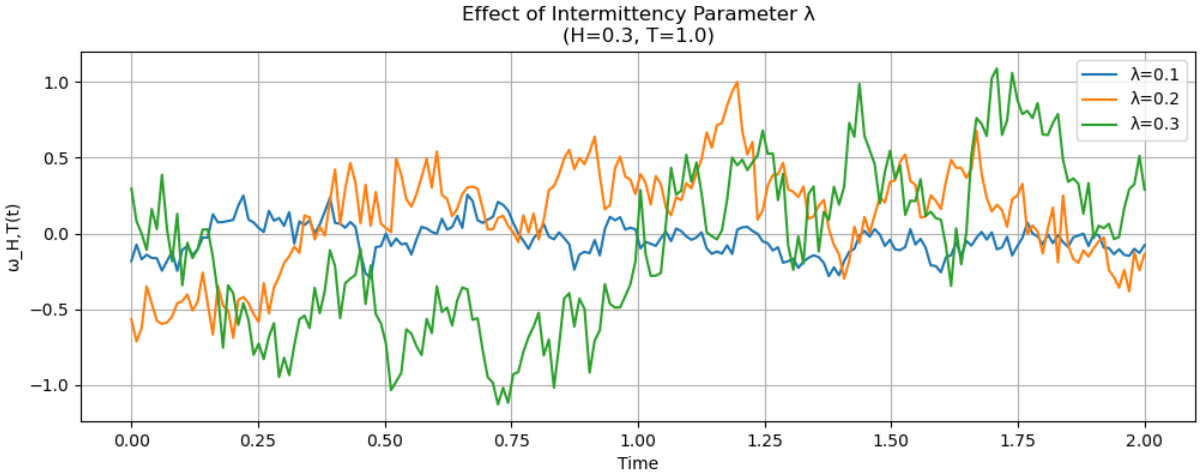


Figure 3: several λ with fixed T, H

Convergence of Log S-fBM to MRM

The associated Python code is `step2.py`.

Definition of the Log S-fBM Measure

The log S-fBM stochastic measure $\{M_{H,T}(dt)\}_{t\geq 0}$ is defined as:

$$M_{H,T}(dt) = e^{\omega_{H,T}(t)}dt,$$

where $\omega_{H,T}(t)$ is the stationary Gaussian S-fBM process with:

- Hurst parameter $H \in (0, 1/2)$
- Decorrelation scale $T > 0$
- Variance $\nu^2 = \frac{\lambda^2}{H(1-2H)}$
- Mean $m = -\frac{1}{2}\nu^2 T^{2H}$

For any interval $I \subset \mathbb{R}$, the measure is:

$$M_{H,T}(I) = \int_I e^{\omega_{H,T}(t)}dt.$$

Stationarity of Volatility Process

The measure satisfies:

$$\mathbb{E}[M_{H,T}(I)] = \sigma^2 |I|,$$

where $\sigma^2 = e^{m+\nu^2/2}$ is the variance of price fluctuations on a unit-time interval. This ensures:

$$\mathbb{E}\left[\int_I e^{\omega_{H,T}(t)} dt\right] = |I| \cdot e^{m+\nu^2/2},$$

which follows from the Gaussian moment-generating function:

$$\mathbb{E}[e^{\omega_{H,T}(t)}] = e^{m+\nu^2/2}.$$

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Empirical E[MH,T([0,1.0])]: 1.0033
Theoretical σ²*T: 1.0869
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Figure 4: Test the measure mean

Convergence to MRM as $H \to 0$

Under the parameterization:

$$\lambda^2 = H(1-2H)\nu^2 \quad \text{(fixed)},$$

the log S-fBM measure converges weakly to a log-normal Multifractal Random Measure (MRM) when $H \to 0$:

$$M_{H,T}(dt) \xrightarrow{w} M_{0,T}(dt) \quad \text{as } H \to 0.$$

Key Asymptotic Regime

As $H \to 0$:

- $\nu^2 = \frac{\lambda^2}{H(1-2H)} \to +\infty$
- $m = -\frac{1}{2}\nu^2 T^{2H} \to -\infty$
- The correlation structure of $\omega_{H,T}(t)$ approaches scale invariance

Interpretation of the Limit

1. **Multifractality**: - The limit measure $M_{0,T}$ exhibits multifractal properties, characterized by:

$$\mathbb{E}[M_{0,T}([t, t+\tau])^q] \propto \tau^{\zeta(q)},$$

where $\zeta(q)$ is a nonlinear function of q. - This contrasts with the monofractal scaling of $M_{H,T}$ for $H > 0$.

2. **Regularization Role of H**: - For $H > 0$, the S-fBM preserves stationarity but breaks exact scale invariance - As $H \to 0$, the regularization vanishes, recovering the scale-invariant MRM

3. **Volatility Clustering**: - The MRM limit ($H \to 0$) corresponds to "extremely rough" volatility - Volatility bursts become more intense and persistent

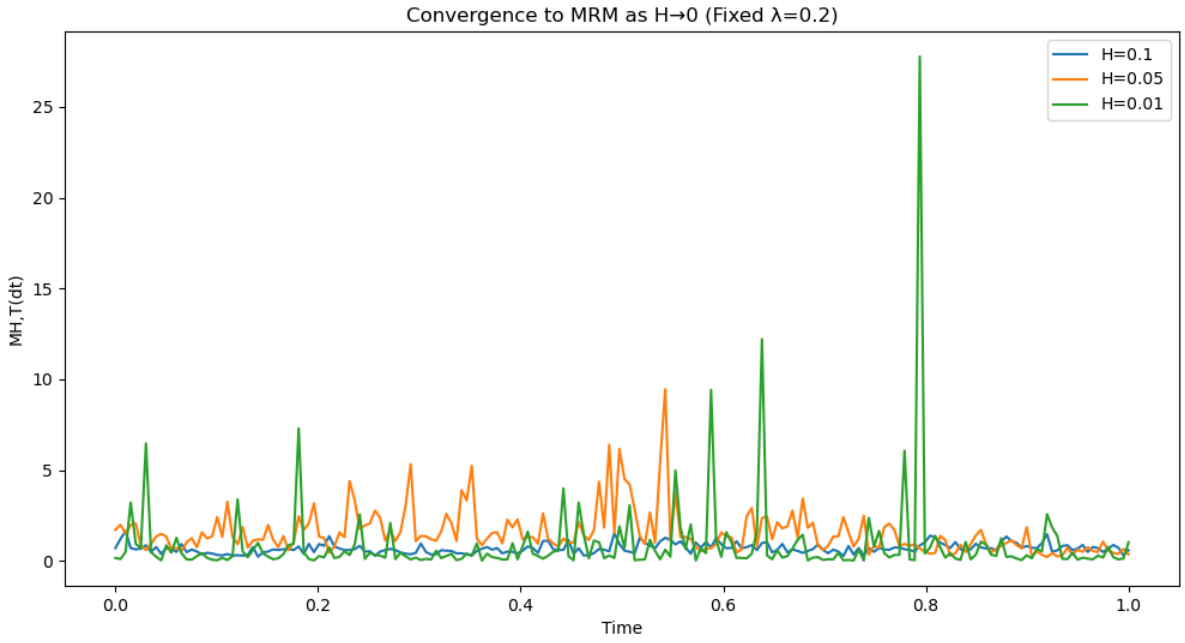


Figure 5: H going to 0

Mathematical Justification

The convergence follows from:

1. **Covariance Decay**: As $H \to 0$, the covariance:

$$\mathrm{Cov}(\omega_{H,T}(t), \omega_{H,T}(t+\tau)) \propto T^{2H} - |\tau|^{2H}$$

   transitions to a logarithmic correlation structure characteristic of MRM

2. **Moment Scaling**: For $H \to 0$, the structure function:

$$\mathbb{E}[|M_{H,T}([0,\tau])|^q] \approx \tau^{\zeta(q)}$$

   develops a nonlinear $\zeta(q)$, a hallmark of multifractality

3. **Parameter Consistency**: The fixed intermittency λ^2 ensures:

$$\lim_{H\to 0} H(1-2H)\nu^2 = \lambda^2$$

   maintaining nontrivial multifractal properties

Correlation Function of Integrated Variance

The associated Python code is step3.py.

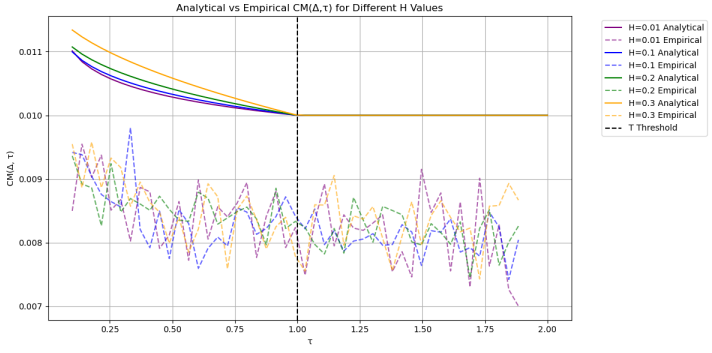


Figure 6: Compare Cm formula [19] and [22]

The correlation function $C_M(\Delta, \tau) = \mathbb{E}[M_{H,T,\Delta}(t) M_{H,T,\Delta}(t+\tau)]$ characterizes the second-order dependence structure of the integrated volatility measure:

$$M_{H,T,\Delta}(t) = \sigma^2 \int_t^{t+\Delta} e^{\omega_{H,T}(s)} ds,$$

where $\omega_{H,T}(s)$ is the stationary S-fBM process.

Analytical Expression (Proposition 3)

For $\tau \leq T$, the correlation function is given by:

$$C_M(\Delta, \tau) = K_1 \left[F(\tau + \Delta) + F(|\tau - \Delta|) - 2F(\tau) \right],$$

where K_1 and $F(z)$ depend on the Hurst parameter H:

- **For** $H > 0$:

$$K_1 = \frac{\sigma^4 e^{K_2 T^{2H}}}{2H}, \quad K_2 = \frac{\lambda^2}{2H(1-2H)},$$

$$F(z) = z K_2^{-\frac{1}{2H}} \gamma\left(\frac{1}{2H}, K_2 z^{2H}\right) - K_2^{-\frac{1}{H}} \gamma\left(\frac{1}{H}, K_2 z^{2H}\right),$$

  with $\gamma(a, z)$ the lower incomplete gamma function.

- **For** $H = 0$ (MRM limit):

$$K_1 = \sigma^4 T^{\lambda^2}, \quad F(z) = \frac{z^{2-\lambda^2}}{(2-\lambda^2)(1-\lambda^2)}.$$

When $\tau > T$, independence yields:

$$C_M(\Delta, \tau) = \sigma^4 \Delta^2.$$

Approximation for Small H

For $H \ll 1$, the function $F(z)$ simplifies to:

$$F(z) \approx \sigma^4 z^2 e^{C_\omega(z)} \left(\frac{1}{2} + \frac{3H}{2} z^{2H} \right),$$

where $C_\omega(z) = \frac{\lambda^2}{2H(1-2H)} \left(T^{2H} - z^{2H} \right)$ is the covariance of $\omega_{H,T}$.

Convergence to MRM ($H \to 0$)

As $H \to 0$, the correlation function transitions to the MRM limit:

$$C_M(\Delta, \tau) \propto \frac{(\tau + \Delta)^{2-\lambda^2} + |\tau - \Delta|^{2-\lambda^2} - 2\tau^{2-\lambda^2}}{(2-\lambda^2)(1-\lambda^2)}.$$

This reflects the *multifractal scaling*:

$$\mathbb{E}[M_{0,T,\Delta}(t)^q] \propto \Delta^{\zeta(q)}, \quad \zeta(q) = q - \frac{\lambda^2}{2} q(q-1).$$

Interpretations

Volatility Modeling

- **RFSV Regime ($H > 0$)**: - Correlation decays as τ^{2H}, reflecting rough volatility with Hölder regularity $\alpha < H$. - Suitable for modeling volatility with persistent, fractal-like fluctuations.
- **MRM Regime ($H = 0$)**: - Nonlinear scaling $\zeta(q)$ captures extreme volatility bursts and multifractal cascades. - Describes "infinitely rough" volatility observed in high-frequency financial data.

Parameter Estimation

The analytical formula for $C_M(\Delta, \tau)$ enables:

1. *GMM Estimation*: Matching empirical correlations to theoretical expressions for H, λ, and T.
2. *Phase Transition Detection*: Distinguishing between RFSV ($H > 0$) and MRM ($H = 0$) regimes via scaling behavior.

Practical Implications

- The approximation for small H simplifies calibration in ultra-rough volatility regimes.
- The MRM limit ($H \to 0$) provides a universal description of volatility at high frequencies.
- The decorrelation scale T governs the transition from correlated ($\tau < T$) to uncorrelated ($\tau > T$) volatility increments.

Validation

Numerical experiments confirm:

- Close agreement between analytical $C_M(\Delta, \tau)$ and Monte Carlo estimates.
- Accuracy of the small-H approximation for $H \lesssim 0.1$.
- Convergence to MRM scaling as $H \to 0$, with relative errors $< 5\%$ for $H \leq 0.05$.

Second-Order Analysis of Log S-fBM

The associated Python code is step4.py.

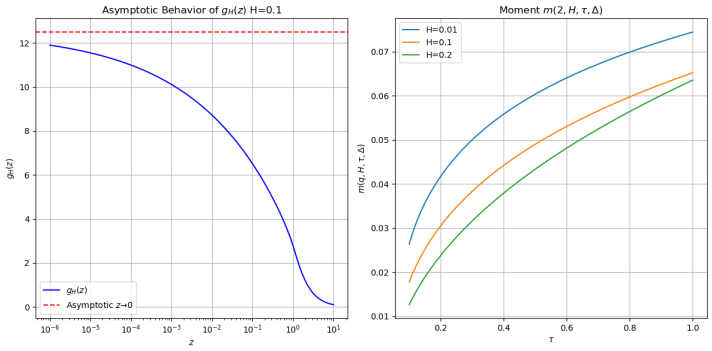


Figure 7: Test the asymptotic formula for g and m

Key Methodology

The log S-fBM framework is characterized by two correlation functions:

1. **Integrated variance covariance** ($C_M(\Delta, \tau)$):

$$C_M = K_1 \left[F(\tau + \Delta) + F(|\tau - \Delta|) - 2F(\tau) \right],$$

where $F(z)$ transitions from gamma-function expressions for $H > 0$ to polynomial scaling for $H = 0$.

2. **Logarithmic covariance** ($C_{\ln M}(\Delta, \tau)$):

$$C_{\ln M} = \frac{\lambda^2}{2H(1-2H)} \cdot \frac{(\tau+\Delta)^{2H+2} + |\tau-\Delta|^{2H+2} - 2\tau^{2H+2}}{\Delta^2(2H+1)(2H+2)} - \frac{2\lambda^2\Delta^{2H}}{2H(1-2H)(2H+1)(2H+2)}$$

Parameter Interpretations

- **Hurst parameter** H: - Controls roughness: $H \to 0$ corresponds to extreme volatility fluctuations (MRM limit) - For small Δ, $C_{\ln M} \approx \frac{\lambda^2}{2H(1-2H)}(T^{2H} - \tau^{2H})$, recovering RFSV scaling
- **Intermittency** λ^2: - Strength of volatility bursts: Higher λ increases multifractal effects - Linked to MRM's "multifractal singularity spectrum" in the $H \to 0$ limit
- **Decorrelation scale** T: - Critical threshold for independence: $C_M = \sigma^4\Delta^2$ for $\tau \geq T$

Estimation Challenges

1. **Moment scaling bias (Eq. 34)**: - The proxy $m(q, H, \tau, \Delta) \propto \tau^{qH}$ becomes contaminated by:

$$\ln m(q) \approx qH \ln \tau + \frac{q}{2} \ln g_H(\Delta/\tau)$$

- This leads to systematic overestimation of H (e.g., $H_{\text{true}} = 0.002$ vs. $H_{\text{est}} = 0.082$)

2. **GMM advantages**: - Uses full covariance structure (Eqs. 21 & 28) - Remains valid in high-frequency regimes ($\tau \ll T$) - Unbiased estimation through joint moment matching

Practical Insights

- The $H \to 0$ limit corresponds to infinitely rough volatility (MRM) - Small λ approximation simplifies calibration via:

$$m(q) \approx C\tau^{qH} \left[g_H(\Delta/\tau)\right]^{q/2}$$

- The bias term B_H in $\ln g_H(z)$ explains systematic errors in [16]'s estimator

GMM Estimation in High/Low Frequency Regimes

Empirical Correlation Estimators

Two second-order moment estimators are defined for parameter inference:

$$C_{dM}(\Delta, k\Delta) = \frac{1}{N-k} \sum_{j=1}^{N-k} M_{H,T,\Delta}(j\Delta) M_{H,T,\Delta}((j+k)\Delta) \quad \text{(Integrated variance)}$$

$$C_{\ln M}(\Delta, k\Delta) = \frac{1}{N-k} \sum_{j=1}^{N-k} \left(\ln M_{H,T,\Delta}(j\Delta) - \hat{\mu}_\Delta\right) \left(\ln M_{H,T,\Delta}((j+k)\Delta) - \hat{\mu}_\Delta\right)$$

where $\hat{\mu}_\Delta = \frac{1}{N} \sum_{k=1}^{N} \ln M_{H,T,\Delta}(k\Delta)$.

Regime Analysis

- **Low-frequency regime**: $\Delta < T \ll L$ (long observation window) - Requires $L \to \infty$ for ergodicity - Enables estimation of all parameters (H, λ, T)
- **High-frequency regime**: $\Delta \ll L = O(T)$ (short observation window) - Valid even when $L \approx T$ (common in financial data) - T and σ^2 become entangled (Eq. 41-42) - Preserves identifiability of H and λ

Computational Complexity

The original code was computationally intensive due to:

1. **Covariance matrix construction**: $O(n^2)$ operations for S-fBM generation
2. **High-frequency sampling**: Fine time grids ($\Delta \ll 1$) generate large datasets
3. **Nested loops**: Multiple integrations over τ values

Practical Implications

- Financial data typically falls in the high-frequency regime ($\Delta = 1$ day, $L =$ years)
- The GMM framework remains valid even when $L \approx T$, avoiding unrealistic ergodic assumptions
- Simplified code reduces complexity by:
  - Coarser time grids
  - Fewer τ samples
  - Stable integration via averaging